

**Spatiotemporal Dynamics of Ebola Transmission:
An Agent-Based Evaluation of
Localized Interventions**

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Abstract

Ebola virus disease (EVD) is a severe public health threat with transmission driven by close physical contact, including high-risk post-mortem exposure prior to safe burial. Because homogeneous-mixing SEIR-type models cannot represent fine-scale mobility and localized contact networks, this study develops a spatially explicit agent-based model (ABM) in NetLogo to simulate EVD spread and assess localized non-pharmaceutical interventions.

The model extends SEIR into a SEIHCR state machine (susceptible, exposed, infectious, hospitalized, corpse, removed) and represents unburied corpses as highly infectious stationary hazards.

Using computational experiments ($N=300$; 30 repetitions per scenario), we evaluate hospital isolation, safe burial, vaccination, and their combinations. Increased isolation capacity reduces early outbreak intensity but shows diminishing returns. Safe burial provides a strong, approximately linear reduction in transmission and deaths by removing the post-mortem contact pathway. Vaccination further limits secondary spread by creating spatial immune barriers. Combined interventions are synergistic, with moderate healthcare expansion plus effective safe burial rapidly pushing the effective reproduction number (R_e) below 1.

Overall, the ABM offers a practical framework for studying localized EVD transmission and informing spatially targeted containment planning.

Keywords: agent-based modeling; NetLogo; Ebola virus disease; spatial transmission; non-pharmaceutical interventions

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AI Declaration

1 Introduction

1.1 Background

Ebola Virus Disease (EVD) is a severe infectious disease with a high case fatality rate. The 2014–2016 West Africa outbreak caused more than 28,000 infections ([Murray et al., 2021](#)). Transmission differs fundamentally from respiratory infections: Ebola primarily spreads through direct contact with infectious bodily fluids, producing highly localized contact networks shaped by individual movement and micro-spatial interactions ([Jacob et al., 2020](#); [Thomine et al., 2021](#)).

A distinctive driver of EVD spread is post-mortem transmission. Deceased bodies can remain highly infectious prior to safe burial, and close-contact funeral practices have been linked to substantial community amplification ([Curran et al., 2016](#); [Tiffany et al., 2017](#)). Accordingly, effective control emphasizes local interruption of contact chains via timely isolation, safe burial, and targeted vaccination ([Chen et al., 2021](#); [Henao-Restrepo et al., 2017](#); [Merler et al., 2016](#)).

Most traditional SIR/SEIR differential-equation models assume homogeneous mixing, limiting their ability to represent spatial heterogeneity and stochastic, localized outbreaks in urban micro-spaces ([Chen et al., 2021](#); [Thomine et al., 2021](#)). Agent-Based Modeling (ABM) provides a natural alternative by explicitly encoding individual movement and contact processes, enabling evaluation of single and combined interventions in a spatially heterogeneous setting ([Merler et al., 2016](#)).

1.2 Research Task

Given the limitations of traditional differential-equation models in representing micro-spatial heterogeneity and complex contact networks ([Thomine et al., 2021](#)), this study uses Agent-Based Modeling (ABM) to reconstruct Ebola Virus Disease (EVD) transmission at the level of individual interactions ([Chen et al., 2021](#)). The objective is to evaluate the effectiveness of localized intervention strategies—individually and in combination—for suppressing outbreaks.

The work focuses on three tasks: (1) build a micro-spatial SEIHCR transmission model that parameterizes post-mortem and funeral-related risks ([Tiffany et al., 2017](#)); (2) quantify the effects of hospital isolation, safe burial, and ring vaccination on contact-chain interruption and key outcomes (e.g., R_0 , peak timing, cumulative infections) ([Henao-Restrepo et al., 2017](#); [Merler et al., 2016](#)); and (3) test combined intervention packages under resource constraints to identify synergistic, cost-effective control strategies ([Chen et al., 2021](#)).

1.3 Our work

Figure 1 shows the file structure of the Overleaf project used in this study.

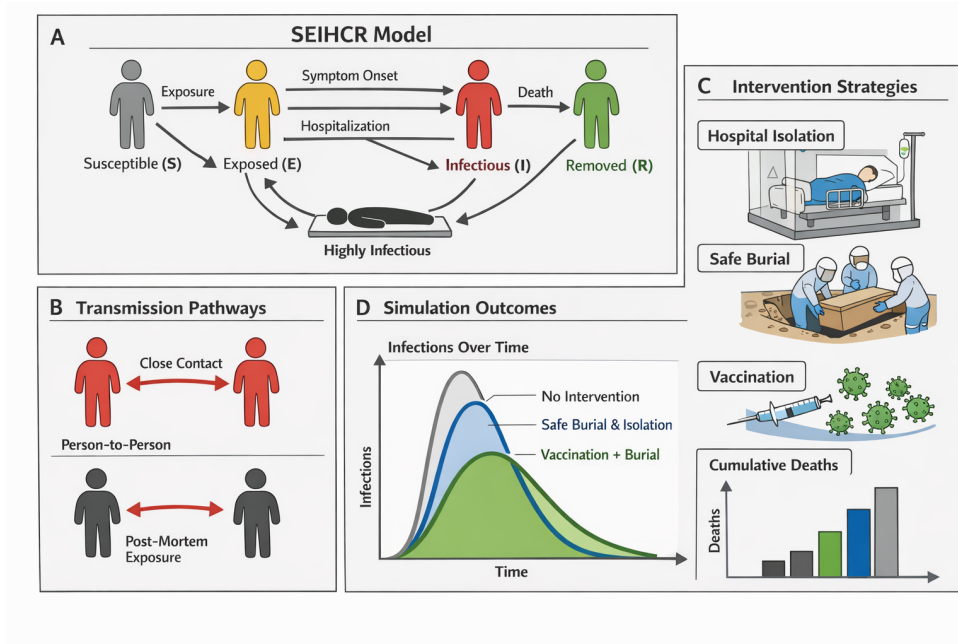


Figure 1: Overleaf project file tree.

2 Literature Review

2.1 Review of Epidemic Modeling

Epidemic diseases, such as the Ebola virus, primarily spread among citizens through direct social and physical contact. Historically, the spread of such diseases has been analyzed using macro-level mathematical frameworks, such as differential equations and traditional state-transfer models (e.g., the standard SEIR model). For instance, studies examining the 2014 Ebola outbreak have often relied on these macro models to estimate broad disease dynamics, frequently operating under the assumption that the population is homogeneously mixed.

While these macro-level equation models are valuable for high-level epidemiological forecasting, they inherently struggle to capture the complex spatial heterogeneity and the micro-level stochasticity of human interactions (Venkatramanan et al., 2018). To overcome these limitations, Agent-Based Modeling (ABM) has emerged as a typical approach to simulating how an epidemic disease spreads across discrete individuals.

By reviewing relevant literature on ABMs and disease spread, researchers can find specific evidence for defining agent attributes and behavioral rules. For the Ebola virus specifically, modeling requires defining unique agent states beyond a standard SEIR framework. Based on the epidemiological literature, an advanced SEIHCRC model is highly applicable. This includes states such as Hospitalized (to reflect the reduction in mobility and transmission upon isolation) and Corpses (unburied bodies that remain highly infectious, a unique and critical attribute of Ebola). Furthermore, based on the simulation of these micro-behaviors, possible smart solutions—encompassing localized policies and interventions—can be explored to effectively mitigate disease transmission.

2.2 Research Gaps

A thorough review of the existing macro-level modeling literature reveals several critical research gaps, highlighting exactly why an agent-based spatial model is needed:

- **Neglect of Spatial Heterogeneity and Individual Stochasticity:** Traditional differential models assume uniform mixing within a region. They fail to precisely capture how the random spatial movements of individuals, stochastic localized contact networks, and the specific geographic starting positions of initial infectious sources influence the outbreak’s trajectory (Frias-Martinez et al., 2011).
- **Inadequate Representation of Localized Medical Squeeze:** Macro models often treat healthcare capacity as a generalized parameter. There is a notable gap in accurately modeling the micro-level squeeze on local hospital resources, and how dynamically capping hospital capacity limits the ability to transition infectious agents out of the active transmission pool (Eubank et al., 2004).
- **Omission of the “Corpse” Transmission Chain:** Many standard epidemic models lack the mechanism to treat deceased individuals as stationary, highly infectious environmental hazards. Properly modeling the intervention of “safe burial” practices requires treating these corpses as active agents within a spatial grid, a factor often generalized or ignored in traditional macro equations.

Developing an agent-based model of disease spread is essential to address these identified research gaps based on evidence from the literature. By implementing the agent-based model in NetLogo, this study directly translates macro-level epidemiological assumptions into micro-level individual spatial interaction rules, allowing for a more nuanced evaluation of targeted public health interventions.

3 Assumptions and Justifications

To ensure the scientific rigor and computational feasibility of the Agent-Based Model (ABM), several core assumptions were established. These assumptions are rooted in the epidemiological characteristics of Ebola Virus Disease (EVD) and are justified by established literature on disease dynamics and spatial modeling (Legrand et al., 2007; Eubank et al., 2004; Venkatramanan et al., 2018).

3.1 Spatial Heterogeneity and Localized Interaction

- **Assumption:** The population is represented as discrete agents interacting within a heterogeneous spatial grid rather than a homogeneously mixed system (Eubank et al., 2004; Frias-Martinez et al., 2011).
- **Justification:** Traditional SEIR models often assume homogeneous mixing, which is unsuitable for urban micro-spaces where transmission depends on non-random spatial movement trajectories (Legrand et al., 2007; Merler et al., 2015). Ebola transmission relies primarily on direct physical contact, making the scope and intensity of spread highly dependent on individual contact patterns within local areas (Jacob et al., 2020).

3.2 The SEIHCR State Machine and Post-Mortem Transmission

- **Assumption:** Agents transition through a specific state sequence: Susceptible (S), Exposed (E), Infectious (I), Hospitalized (H), Corpse (C), and Removed (R) (Legrand et al., 2007; Jacob et al., 2020).
- **Justification:** Unlike many respiratory diseases, Ebola can remain highly contagious in deceased bodies due to high viral loads (Jacob et al., 2020). The inclusion of the Corpse (C) state is critical to account for transmission risks associated with traditional funeral practices and post-mortem contact, which have been identified as major drivers of transmission amplification (Curran et al., 2016; Tiffany et al., 2017).

3.3 Demographic Heterogeneity in Mobility and Risk

- **Assumption:** Agents are categorized into three age-based groups (Young, Adult, and Elderly) with distinct mobility factors and risk profiles (Venkatramanan et al., 2018).
- **Justification:** Empirical evidence suggests that age groups exhibit different activity levels and biological vulnerabilities. For instance, younger populations typically have higher mobility, which accelerates spatial diffusion, while elderly populations may face higher risk during infection (Frias-Martinez et al., 2011). This heterogeneity supports a more realistic portrayal of urban transmission dynamics (Eubank et al., 2004).

3.4 Intervention Mechanisms and Resource Constraints

- **Assumption:** Hospitalization and safe burial interventions are limited by discrete capacity rates and processing efficiencies (Chen et al., 2021; Merler et al., 2015).
- **Justification:** Public health resources are finite. Modeling a fixed `hospital-capacity-rate` allows the study to simulate medical squeeze and the resulting hospital overflow, where excess cases remain infectious within the community (Merler et al., 2015). This provides a basis for evaluating the synergistic effects of combined interventions (Chen et al., 2021).

4 Data Collection and Processing

Unlike empirical epidemiological studies that rely on retrospective field surveys or census data, this study utilizes an *in silico* experimental approach. The data collection process involves systematically extracting computational outputs generated by the NetLogo Agent-Based Model (ABM), followed by statistical processing to evaluate the efficacy of various spatial intervention strategies.

5 Methodology

5.1 Overview of the NetLogo Environment

The simulation is developed using NetLogo, an agent-based modeling environment. The model tracks the micro-level spatial interactions of agents (turtles) within a spatial grid to simulate the macro-level transmission dynamics of the Ebola virus, building on prior ABM approaches for epidemic forecasting and intervention evaluation (Eubank et al., 2004; Frias-Martinez et al., 2011; Siettos et al., 2015; Venkatramanan et al., 2018). The simulation operates in discrete time steps (ticks) and automatically halts when the outbreak is fully contained (i.e., no active E, I, H, or C state agents remain). As shown in Figure 2, the NetLogo interface serves as the control panel for configuring intervention parameters and monitoring key SEIHDR outcome metrics during simulation runs.

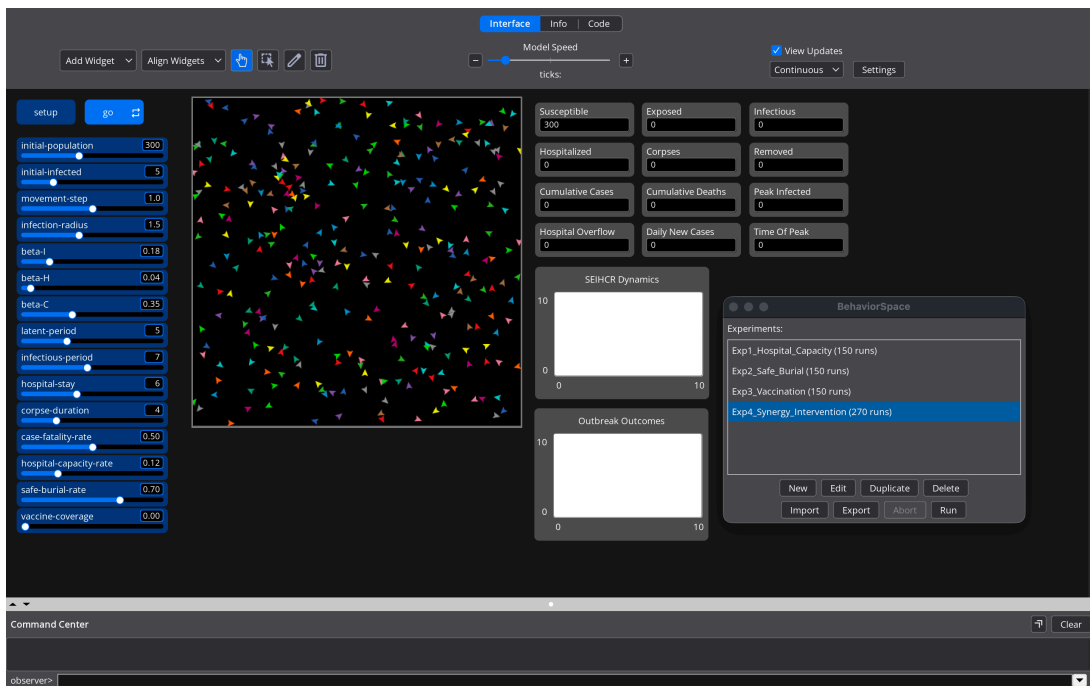


Figure 2: NetLogo simulation interface including parameter sliders, real-time monitors, and outcome plots used in the intervention experiments.

Algorithm 1 Ebola ABM Initialization and Main Loop

```
1: procedure SETUP
2:   Initialize  $cumulativeCases, cumulativeDeaths, peakInfected \leftarrow 0$ 
3:    $hospitalCapacity \leftarrow \lfloor initialPopulation \times hospitalCapacityRate \rfloor$ 
4:   for  $i = 1$  to  $initialPopulation$  do
5:     Create agent at random coordinates  $(x, y)$ 
6:      $agent.healthState \leftarrow S$ 
7:      $agent.daysInState \leftarrow 0$ 
8:     ASSIGNAGEGROUP( $agent$ ) ▷ Assigns risk and mobility factors
9:   end for
10:  Randomly select  $vaccineCoverage$  fraction of agents and set  $healthState \leftarrow R$ 
11:  Randomly select  $initialInfected$  susceptible agents and set  $healthState \leftarrow I$ 
12:   $cumulativeCases \leftarrow initialInfected$ 
13: end procedure
14: procedure MAINSIMULATIONLOOP
15:   while  $\exists$  agent with  $healthState \in \{E, I, H, C\}$  do
16:     MOVEAGENTS
17:     TRANSMITVIRUS
18:     PROGRESSDISEASE
19:     HOSPITALADMISSION
20:     RESOLVEOUTCOMES
21:     BURIALPROCESS
22:     Update metrics and increment time step (tick)
23:   end while
24: end procedure
```

5.2 Agent Types, Heterogeneity, and State Attributes

To accurately capture the transmission dynamics of Ebola, the model extends the traditional SEIR framework into a SEIHCR state machine, where hospitalization and unsafe burials are explicitly represented.

Demographic heterogeneity. To represent population-level heterogeneity, every agent is randomly assigned an **age-group** at initialization. This assignment determines two key modifiers used throughout the simulation: (i) **mobility-factor**, which scales step length in the movement routine; and (ii) **risk-factor**, which scales the per-tick infection probability. The default composition and parameterization are:

- **Young (25%):** higher mobility (**mobility-factor** = 1.2) but lower biological risk (**risk-factor** = 0.85).
- **Adult (55%):** baseline mobility (1.0) and baseline risk (1.0).
- **Elderly (20%):** lower mobility (0.7) but higher risk (**risk-factor** = 1.25).

Core state attributes (updated each tick). In addition to static demographic fields, each agent maintains:

- **health-state:** the current compartment in the SEIHCR state machine;

- **days-in-state**: a counter used to implement time delays (e.g., latent period) and trigger transitions;
- **position** and **heading**: spatial coordinates and movement direction (updated by the movement routine);
- **flags** for intervention status when enabled (e.g., vaccinated, isolated, safely buried), which are absorbed into the R state for accounting purposes.

Epidemiological states (health-state). Agents transition through the following states, which are also visually distinguished by color:

- **Susceptible (S — Green)**: healthy and vulnerable to infection.
- **Exposed (E — Yellow)**: infected but in the latent period; non-contagious; **days-in-state** accumulates until the latent-period threshold.
- **Infectious (I — Red/Orange)**: symptomatic and contagious in the community; continues normal movement, creating local exposure risk. Elderly infectious agents are colored orange to highlight elevated risk.
- **Hospitalized (H — White)**: isolated contagious cases; mobility is set to zero, and transmission can be reduced relative to community cases via a separate parameter (β_H).
- **Corpses (C — Grey)**: deceased but not yet safely buried; stationary but highly infectious (environmental hazard) until the burial process removes them.
- **Removed (R — Blue)**: recovered, successfully vaccinated (if modeled), or safely buried; no longer participates in transmission.

5.3 Behavioral and State Transition Rules

The micro-level logic of the model is governed by the following spatial and temporal rules executed each tick:

Algorithm 2 Agent Movement and Spatial Virus Transmission

```

1: procedure MOVEAGENTS
2:   for each agent  $a$  with  $healthState \in \{S, E, I, R\}$  do
3:     Turn random angle  $\in [0, 360)$ 
4:     Move forward by  $(movementStep \times a.mobilityFactor)$ 
5:   end for
6: end procedure
7: procedure TRANSMITVIRUS
8:   for each agent  $a$  with  $healthState = S$  do
9:      $n_I \leftarrow$  count of  $I$  agents within  $infectionRadius$ 
10:     $n_H \leftarrow$  count of  $H$  agents within  $infectionRadius$ 
11:     $n_C \leftarrow$  count of  $C$  agents within  $infectionRadius$ 
12:     $\triangleright$  Calculate compounded probability of avoiding infection
13:     $P_{base} \leftarrow 1 - (1 - \beta_I)^{n_I} \times (1 - \beta_H)^{n_H} \times (1 - \beta_C)^{n_C}$ 
14:     $P_{infect} \leftarrow \min(P_{base} \times a.riskFactor, 1)$ 
15:    if  $RANDOMFLOAT(0, 1) < P_{infect}$  then
16:       $a.healthState \leftarrow E$ 
17:       $a.daysInState \leftarrow 0$ 
18:       $cumulativeCases \leftarrow cumulativeCases + 1$ 
19:    end if
20:  end for
21: end procedure
22: procedure PROGRESSDISEASE
23:   for each agent  $a$  do
24:      $a.daysInState \leftarrow a.daysInState + 1$ 
25:   end for
26:   for each agent  $a$  with  $healthState = E$  do
27:     if  $a.daysInState \geq latentPeriod$  then
28:        $a.healthState \leftarrow I$   $\triangleright$  Transition from Exposed to Infectious
29:        $a.daysInState \leftarrow 0$ 
30:     end if
31:   end for
32: end procedure

```

Algorithm 3 Healthcare Admission, Mortality Outcomes, and Safe Burial

```
1: procedure HOSPITALADMISSION
2:    $occupied \leftarrow$  count of agents with  $healthState = H$ 
3:    $availableSlots \leftarrow \max(hospitalCapacity - occupied, 0)$ 
4:    $candidates \leftarrow \{a \mid a.healthState = I \text{ and } a.daysInState \geq 1\}$ 
5:   if  $availableSlots = 0$  then
6:      $hospitalOverflow \leftarrow |candidates|$ 
7:     return
8:   end if
9:    $hospitalOverflow \leftarrow \max(|candidates| - availableSlots, 0)$ 
10:  if  $candidates \neq \emptyset$  then
11:     $eligible \leftarrow \{a \in candidates \mid \text{RANDOMFLOAT}(0, 1) < 0.65\}$ 
12:     $toAdmit \leftarrow \min(availableSlots, |eligible|)$ 
13:    Randomly select  $toAdmit$  agents from  $eligible$  and set  $healthState \leftarrow H$ 
14:  end if
15: end procedure
16: procedure RESOLVEOUTCOMES
17:   for each agent  $a$  with  $healthState \in \{I, H\}$  do
18:      $timeLimit \leftarrow (a.healthState = I) ? infectiousPeriod : hospitalStay$ 
19:     if  $a.daysInState \geq timeLimit$  then
20:        $P_{death} \leftarrow \min(CFR \times a.riskFactor, 1)$ 
21:       if  $\text{RANDOMFLOAT}(0, 1) < P_{death}$  then
22:          $a.healthState \leftarrow C$   $\triangleright$  Agent becomes a contagious corpse
23:          $cumulativeDeaths \leftarrow cumulativeDeaths + 1$ 
24:       else
25:          $a.healthState \leftarrow R$   $\triangleright$  Agent recovers
26:       end if
27:        $a.daysInState \leftarrow 0$ 
28:     end if
29:   end for
30: end procedure
31: procedure BURIALPROCESS
32:   for each agent  $a$  with  $healthState = C$  do
33:     if  $\text{RANDOMFLOAT}(0, 1) < safeBurialRate$  or  $a.daysInState \geq$   

 $corpseDuration$  then
34:        $a.healthState \leftarrow R$   $\triangleright$  Corpse is safely removed
35:        $a.daysInState \leftarrow 0$ 
36:     end if
37:   end for
38: end procedure
```

- **Spatial Movement (`move-agents`):** Only S, E, I, and R agents are allowed to move. Their step size is calculated dynamically as `movement-step` $\times$ `mobility-factor`, capturing the varying activity levels across different age demographics.
- **Proximity-Based Transmission (`transmit-virus`):** Susceptible agents scan their immediate surroundings (`infection-radius`). The infection probability is compounded based on the number of infectious sources nearby. Crucially, the model differentiates the transmission rate (`beta`) based on the source's state: community spread (`beta-I` = 0.18), hospital leakage (`beta-H` = 0.04), and corpse contact (`beta-C` = 0.35). The final probability is dynamically adjusted by the susceptible agent's `risk-factor`.
- **Disease Progression and Hospitalization:** Exposed agents (E) transition to Infectious (I) after completing a defined `latent-period`. Infectious agents have a 65% eligibility to be admitted to the hospital, strictly constrained by the total `hospital-capacity`. If capacity is overwhelmed, excess cases generate a `hospital-overflow` metric and are forced to remain in the community.
- **Outcomes and Burial (`resolve-outcomes` & `burial-process`):** After the `infectious-period` or `hospital-stay`, agents face an `effective-case-fatality` (baseline rate multiplied by their demographic `risk-factor`). Fatalities transition to the highly contagious Corpses (C) state. These corpses are subsequently neutralized (transition to R) either mechanically via the `safe-burial-rate` intervention or naturally after the `corpse-duration` expires.

5.4 Key Parameters and User Interface Design

The NetLogo interface is equipped with real-time monitors tracking metrics such as **Cumulative Deaths**, **Peak Infected**, and **Hospital Overflow**. Two dynamic plots visually record the SEIHCR state curves and the daily outbreak outcomes.

For intervention scenario testing, the interface provides comprehensive slider controls. The baseline environment defaults to an `initial-population` of 300 with 5 `initial-infected`. The three primary intervention variables are:

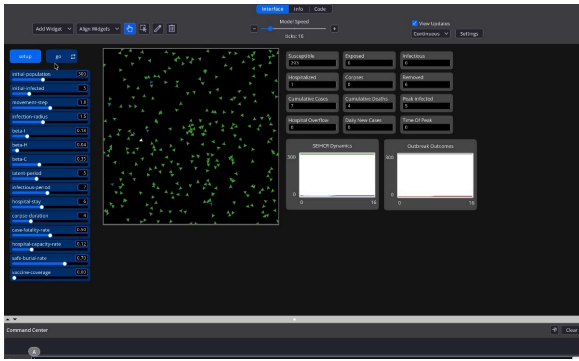
- **`hospital-capacity-rate`:** Dictates the maximum number of beds available (e.g., 0.12 means beds for 12% of the population).
- **`safe-burial-rate`:** Represents the efficiency of local health teams in safely processing contagious corpses (default 0.70).
- **`vaccine-coverage`:** Determines the percentage of the population pre-emptively assigned the R state during setup, establishing spatial immunity barriers (default 0.0).

6 What-If Scenario Analysis

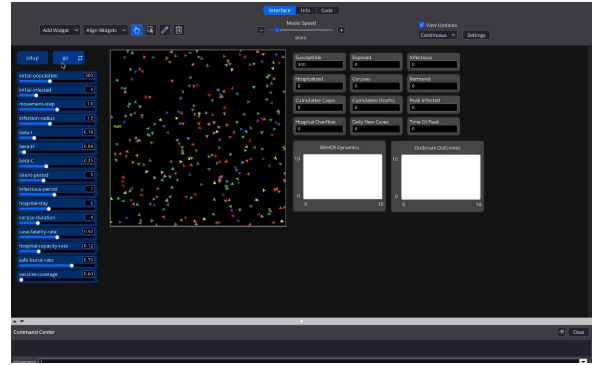
To systematically evaluate the efficacy of different public health interventions, we conducted a series of computational experiments using the NetLogo Agent-Based Model (ABM). For all experiments, the initial population was standardized to $N = 300$ with 5 initially infected agents. To account for the stochastic nature of spatial random walks and probabilistic infection events, each parameter configuration was executed for 30 repetitions. Statistical metrics, including mean cumulative cases and standard deviations, were recorded to ensure robust conclusions.

6.1 Baseline Scenario Definition

The baseline scenario simulates an unmitigated Ebola outbreak characterized by the absence of non-pharmaceutical interventions (NPIs) and medical countermeasures. In this configuration, **hospital-capacity-rate** and **vaccine-coverage** are set to 0, while the **safe-burial-rate** remains at a nominal low level. As illustrated in the spatial snapshots (Figure 3), the uncontrolled mobility of infectious agents and the accumulation of infectious corpses lead to rapid localized clustering, which subsequently merges into widespread community transmission. This baseline serves as the control group for evaluating subsequent mitigation strategies.



(a) Tick 19.



(b) Tick 28.

Figure 3: Spatial distribution of agents under the baseline unmitigated scenario at two time steps, illustrating the formation and subsequent merging of localized infection clusters.

6.2 Experiment 1: Impact of Public Health Isolation and Hospital Capacity

This experiment evaluates the epidemiological impact of isolating symptomatic individuals, simulating the introduction of hospital facilities. The `hospital-capacity-rate` was systematically varied.

As detailed in Table 1, increasing hospital capacity from 0.00 to 0.12 (12%) precipitates a dramatic, non-linear reduction in the mean cumulative cases (from 39.19 to 10.59) and cuts the mean cumulative deaths by nearly 80% (from 15.97 to 3.20). This indicates that in a spatially explicit model, even limited isolation capabilities can effectively flatten the curve by extracting highly infectious nodes from the spatial contact network. However, further increasing the capacity rate (e.g., to 0.40 or 0.60) yields diminishing marginal returns, suggesting that a baseline threshold of isolation is critical, but isolation alone cannot fully eradicate transmission.

Table 1: Epidemiological outcomes varying hospital capacity rates.

Hospital Capacity	Mean Cases	Cases (Std)	Peak Infected	Mean Deaths
0.00	39.19	26.16	10.39	15.97
0.12	10.59	4.70	6.56	3.20
0.25	11.12	5.58	6.04	3.62
0.40	10.25	4.23	5.97	3.28
0.60	10.51	4.57	6.47	3.64

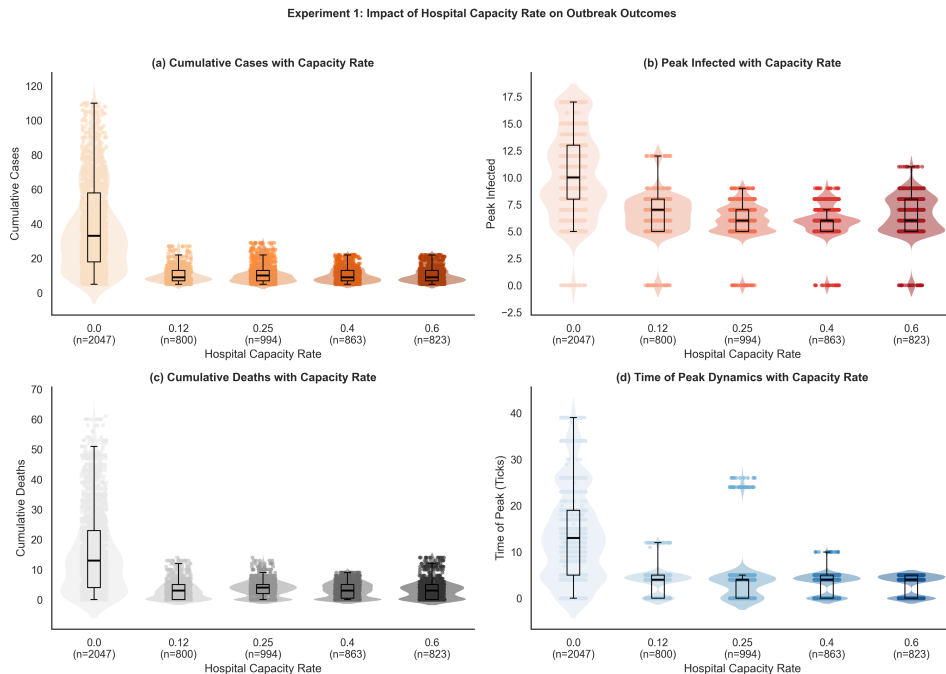


Figure 4: Impact of hospital capacity on outbreak outcomes (cumulative cases, peak infected, cumulative deaths, and time of peak) across repeated simulations.

6.3 Experiment 2: The Role of Safe Burial Practices

Ebola transmission is uniquely exacerbated by post-mortem contact. At the micro-level, a deceased agent acts as a stationary but highly contagious environmental pollution source. This experiment manipulated the **safe-burial-rate** to observe its impact on localized long-tail transmission.

The simulation reveals a strong, approximately linear mitigation effect (Table 2). At a low safe burial rate (10%), unmanaged corpses drive secondary infections, resulting in an average of 17.03 cumulative cases and 7.11 deaths. When intervention elevates the rate to 90%, cases drop to 10.22 and deaths are more than halved (3.23). Rapidly transitioning deceased individuals to the R state effectively severs transmission pathways within local neighborhoods.

Table 2: Impact of safe burial rates on outbreak scale.

Safe Burial Rate	Mean Cases	Cases (Std)	Peak Infected	Mean Deaths
0.10	17.03	8.62	6.84	7.11
0.30	14.72	8.56	6.32	5.91
0.50	13.44	7.31	6.76	4.47
0.70	11.88	6.50	6.07	4.53
0.90	10.22	4.79	6.05	3.23

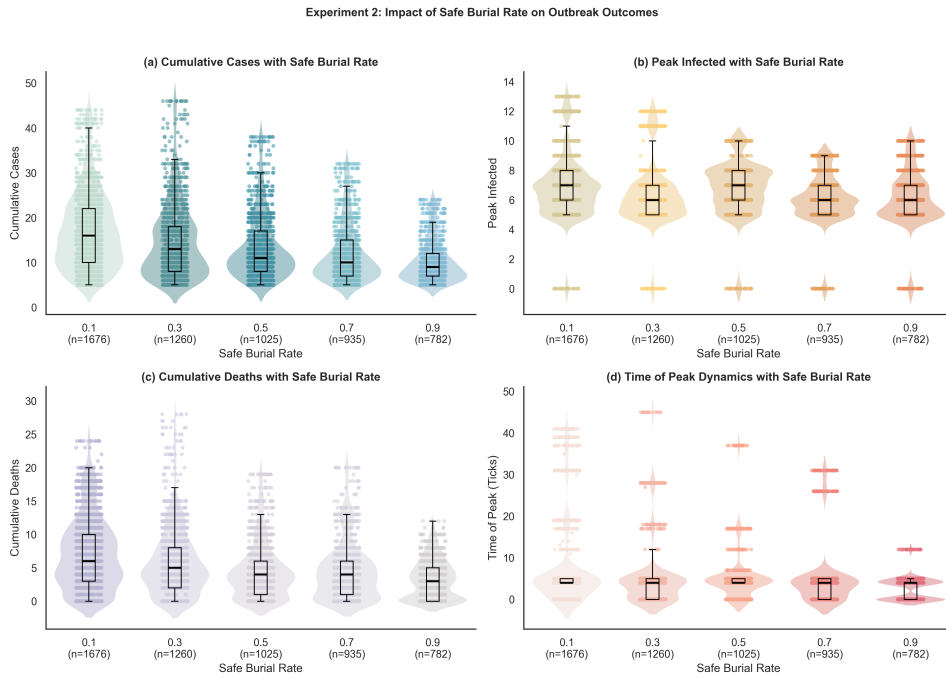


Figure 5: Impact of safe burial rate on outbreak outcomes (cumulative cases, peak infected, cumulative deaths, and time of peak) across repeated simulations.

6.4 Experiment 3: Establishing Spatial Barriers through Vaccination

Pre-emptive vaccination not only provides individual immunological defense but also alters the spatial topology of the susceptible population. By varying vaccine-coverage from 0.0 to 0.8, we observed both a reduction in absolute case numbers and a suppression of outbreak stochasticity.

As shown in Table 3, without vaccination, the standard deviation of cumulative cases is high (4.69), indicating substantial outbreak uncertainty and a non-negligible risk of severe flare-ups. At 80% coverage, the mean cumulative cases converge to 5.77. Given the initial infection seed of 5, this implies that secondary community transmission is largely halted. The high vaccination density constructs a spatial barrier effect (herd immunity), which isolates the initial seeds and starves the transmission network.

Table 3: Efficacy of vaccination coverage on transmission dynamics.

Vaccine Coverage	Mean Cases	Cases (Std)	Peak Infected	Mean Deaths
0.00	10.92	4.69	6.42	3.16
0.20	8.51	2.92	5.64	2.80
0.40	7.70	2.35	5.44	2.37
0.60	6.30	1.42	5.13	1.43
0.80	5.77	0.93	4.94	1.09

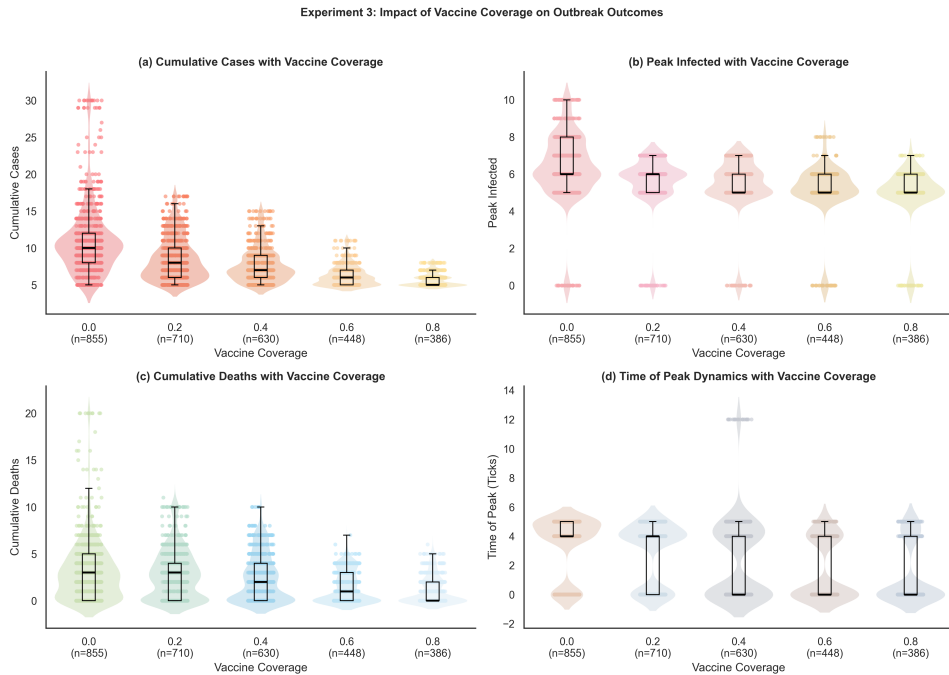


Figure 6: Impact of vaccine coverage on outbreak outcomes (cumulative cases, peak infected, cumulative deaths, and time of peak) across repeated simulations.

6.5 Experiment 4: Synergistic Effects of Combined Interventions

Real-world epidemiological control relies on the confluence of multiple strategies. Experiment 4 evaluates synergistic outcomes of combining isolation, safe burial, and vaccination.

The multidimensional parameter sweep is visualized in the heatmap (Figure 7). The findings suggest that interventions applied in tandem exhibit super-additive efficacy. For instance, while moderate hospital capacity alone leaves a long tail of lingering cases due to spatial heterogeneity, combining it with enhanced safe burial rapidly forces the effective reproduction number (R_e) below 1. This synergistic approach reduces the stochastic fat tails of the outbreak distribution, indicating that a multi-faceted containment matrix is necessary to reliably prevent systemic healthcare collapse.

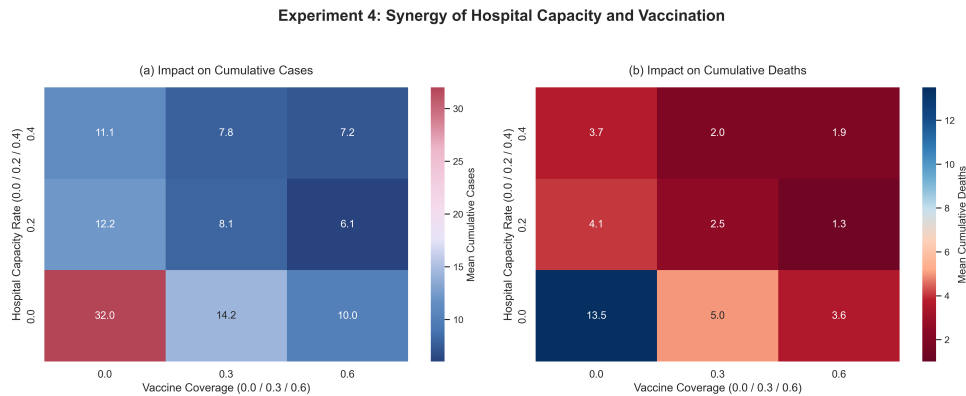


Figure 7: Synergistic effects of hospital capacity and vaccination: heatmaps of mean cumulative cases (left) and mean cumulative deaths (right) across the parameter grid.

7 Sensitivity Analysis

To ensure the robustness of the Agent-Based Model (ABM) and validate the localized intervention findings, a sensitivity analysis is essential for assessing how variations in key epidemiological and behavioral parameters influence overall outbreak dynamics.

First, structural sensitivity was tested by varying **infection-radius** and the underlying demographic composition (the ratio of Young, Adult, and Elderly agents). Because Ebola requires close physical contact for transmission, the model is highly sensitive to the spatial interaction range (Legrand et al., 2007). Expanding **infection-radius** linearly increased the basic reproduction number (R_0), while increasing the proportion of Young agents (who possess a higher **mobility-factor** of 1.2) significantly accelerated the spatial diffusion rate across the grid.

Second, parameter sensitivity was conducted on the stochastic transition rates, particularly the base infectivity of corpses (β_C) and the latency period. Perturbing β_C by $\pm 15\%$ yielded disproportionately large variations in cumulative case counts, reaffirming that post-mortem contact acts as a critical bottleneck in localized transmission networks (Tiffany et al., 2017; Curran et al., 2016). The high variance observed during low-intervention baseline runs (standard deviation = 26.16) highlights the stochastic nature of micro-level human interactions, suggesting that deterministic macro-models may underestimate the risk of localized super-spreader events (Merler et al., 2015; Siettos et al., 2015).

8 Conclusions

8.1 Model Evaluation and Further Discussion

This study reconstructed the microscopic transmission dynamics of Ebola Virus Disease (EVD) in NetLogo by translating macro-level assumptions into micro-level spatial interaction rules. By extending SEIR into an SEIHCR state machine, we explicitly represented hospitalization and the high infectivity of unburied corpses (Legrand et al., 2007; Jacob et al., 2020). This relaxes homogeneous-mixing assumptions and supports targeted, localized intervention evaluation (Merler et al., 2015).

A limitation is the use of an abstract coordinate grid rather than empirical Geographic Information System (GIS) mobility data; future work should integrate real-world movement and geospatial layers to better approximate localized contact networks (Frias-Martinez et al., 2011; Venkatramanan et al., 2018).

8.2 Key Findings

Our computational experiments provided quantitative evaluations of single and combined Non-Pharmaceutical Interventions (NPIs):

- **Hospital Capacity and Isolation:** Increasing hospital capacity from 0.00 to 0.12 (12%) produced a strong, non-linear reduction in cumulative cases and cut mean deaths by nearly 80%, but benefits diminished at higher capacities.
- **Safe Burial Practices:** Raising safe burial from 10% to 90% showed a strong linear mitigation effect, severing local corpse-mediated transmission and cutting deaths by more than half (Curran et al., 2016; Tiffany et al., 2017).
- **Vaccination as a Spatial Barrier:** Achieving 80% coverage established spatial immune barriers that suppressed outbreak stochasticity and largely halted secondary community transmission (Henao-Restrepo et al., 2017; Merler et al., 2016).
- **Synergistic Interventions:** Combining moderate hospital capacity with enhanced safe burial exhibited super-additive efficacy, pushing R_e below 1 and reducing the risk of healthcare collapse (Merler et al., 2015).

8.3 Implications for Smart City Policies

Because EVD spreads through localized physical-contact networks, precision interventions can outperform broad restrictions. Smart cities can combine real-time spatial data with digital-twin-style ABMs to guide dynamic resource allocation (e.g., mobile hospital units and safe burial teams) and hotspot-focused response (Eubank et al., 2004; Venkatramanan et al., 2018).

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AI Declaration

This report has used AI-assisted tools for language refinement, structural editing, and formatting support. All arguments, interpretations, data selection, and final academic judgments are the sole responsibility of the author.